

Assignment #3

Assignment #3 (data analysis)

https://colab.research.google.com/drive/1SB_IMwqNoiMZUZ7KqQ1BkoZ32WcxRdf2

You need a **Google account** for this assignment, permanent or temporary.

In less than 500 words, explain and compare **the epsilon ranges** that you found in part 1 and part 2, then reflect on your understanding of adversarial attacks on deep learning models and their potential implications for society. Include **screenshots** as requested!

Submit pdf file of your report to Canvas.

- **Question 1:** For the image you considered, which method ("white-box" or "black-box") resulted in a more efficient adversarial attack?
- **Question 2:** How could the vulnerability of deep learning models to adversarial attacks impact the trust and reliability of AI systems in critical applications such as healthcare, autonomous driving, and security?
- **Question 3:** What measures can be taken to mitigate these risks and ensure the safe deployment of AI technologies in society?

Deadline: 17 November, Monday, 11:59 pm

Marking individual report (data analysis)

	Weight	0-49%	50-64%	65-74%	75-84%	85-100%	Mark
Structure and organisation	30%	No clear or coherent structure or organisation	Poor or incomplete structure and organisation	Structure and organisation are partially appropriate	Structure and organisation are appropriate	Structure and organisation are appropriate and effective	
Data analysis	50%	Significantly incomplete results. Societal impact and limitations described inadequately or mis-identified	Partially complete results. Societal impact and limitations described incompletely.	Complete results. Societal impact and limitations described, but without sufficient detail, structure or argumentation.	Complete results. Societal impact and limitations described in sufficient detail, but without strong argumentation.	Complete results. Societal impact and limitations described adequately, with sufficient detail, structure and argumentation.	
Written description	20%	Lack of fluency in expression makes it difficult to comprehend	Lack of fluency in expression but still relatively easy to follow	Good level of fluency and clarity in expression and evidence of logical progression of ideas	High level of fluency and clarity in expression and evidence of logical progression of ideas	Sophistication in fluency of expression	
Total	100%						

In accordance with University policy, penalties apply when work is submitted after 11:59 pm on the due date (Sydney time):

- deduction of 5% of the maximum mark for each calendar day after the due date
- after ten calendar days late, a mark of zero will be awarded
- text matching software (Turnitin) will be used for plagiarism detection